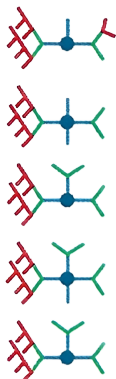


Abstract

This project investigates an unsupervised learning pipeline for the analysis of data recorded from Multi-Electrode Arrays (MEAs), in order to better describe potential data groups that exist. The main independent variable analyzed is the effects bicuculline has on neuron impulse responses; dosages being 0, 1, 3, 5, and 10 μM . To give statistical insights for researchers a deep learning pipeline was implemented, utilizing wavelet based denoising of data which then was passed to a PyTorch based autoencoder to compress large waveforms into smaller dimensional data while preserving signal integrity. The model incorporated Sobolev regularization to keep signal quality while allowing for more uniformity and reducing uncertainties in data. This was then passed into a novel k-means clustering function. The model. The training had a loss of 0.0405, and an average derivative correlation of 0.9907 and kept very good similarity while reducing the overall data to be more easily interpretable. The clusters however were not well defined do to used data, bicuculline was shown not to effect overall voltage response.

Problem Definition

The electro-mechanical behavior of neurons is vital in understanding conditions such as dementia and epilepsy. At Rutgers, researchers use MEAs to record neuron voltage responses under differing conditions, such as different dosages of a neuron GABA inhibitor bicuculline. The researchers only use statistical averaging to compare data, given that a single test has almost 500k data points. However, biologically, what is happening is that under different amounts of this drug, neuron morphology changes. Clustering of this data can give specific insight on possible neuron morphology changes across drug dosages and or genetic perturbation. To extract meaningful structures from high-dimensional, noisy neuroelectrical waveform data, a deep learning pipeline can allow for a scalable environment that preserves biologically relevant features.



Materials and Methods

Data Type: Experimental electrophysiological data from MEA recordings

Source: Dr. Firestein Lab (Rutgers University)

Conditions: 5 bicuculline dosages (0, 1, 3, 5, 10 μM) under one genetic background

Samples Used: ~2,500 spike waveforms selected from 64-electrode recordings

File Explanation:

Each waveform represents neuron voltage response at an electrode of a under excitation (mechanical strain) under drug dosage. Data was stored in .mat files, with each file corresponding to a different concentration level.

Input/Output Structure

Input: 64-sample raw voltage waveform (post-denoising)

Output: 32-dimensional latent vector from the autoencoder

Final visualization: 2D PCA projection of latent space

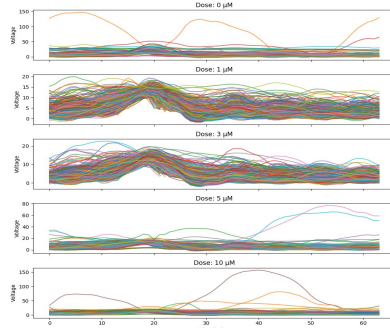
Libraries

PyTorch (autoencoder architecture, Sobolev loss)

NumPy, SciPy (data handling, wavelet transforms)

scikit-learn (PCA, KMeans, clustering metrics)

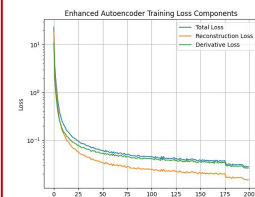
PyWavelets (wavelet-based denoising)



Results

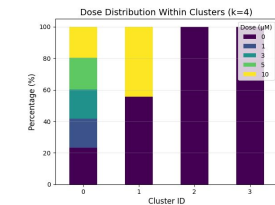
Model and Autoencoder Loss

- Autoencoder Performance
- Final Total Loss: 0.0283
- Reconstruction Loss (MSE): 0.0151
- Sobolev Derivative Loss: 0.0264
- Mean Absolute Difference: 0.0405
- Derivative Correlation: 0.9907



Clustering Metrics

- Silhouette Score: 0.4101
- Davies-Bouldin Index: 0.7988
- Inter-cluster Separation: 2.4107
- Intra-cluster Density: 0.2301



Key Takeaways

- The Sobolev autoencoder learns to preserves shape and slope of waveforms, something traditional PCA or wavelet methods cannot do.
- Embeddings from the model capture signal structure across spikes, allowing for smoother reconstruction even in noisy conditions.
- The model builds representations directly from raw signals, enabling flexibility conditions instead of fitting into a geometric equation

Future Direction and Applications

- Scale to full dataset (~4.8M waveforms) to uncover broader trends
- Better understand k-means clustering and clustering methods asn
- Modularize code for flexible reuse and integration with future datasets
- Enable dose prediction and real-time waveform compression using learned embeddings

Acknowledgments

Thank you to my instructor, Dr. Vlassis and staff of the Applied AI course, as well as to prior collaborators and researchers who collected the MEA data from Dr. Firestein lab at Rutgers